

Software Requirements Specification (SRS)

BEAMS: Brand Exposure Analytics and Metrics System

Group: A03

Members

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1 Introduction

1.1 Overview

Sponsorship is one of the main ways in which brands reach large sports audiences, yet its value is hard to measure. Most existing tools only report how long a logo appears on screen and treat every appearance as equally valuable, even though a logo at the centre of the frame is far more likely to be noticed than one at the edge of a crowded scene. This document outlines the requirements for developing BEAMS, an application that detects and tracks sponsor logos in sports broadcast footage and scores each exposure by the viewer attention it is likely to capture. The results can be queried in plain English.

1.2 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a detailed description of the BEAMS application, including its functionality, user interaction, and performance metrics. This document will serve as a guide for developers, testers, and stakeholders throughout the project lifecycle.

1.3 Brief Description of the Proposed System

The proposed system will detect sponsor logos in each frame of a broadcast video and follow them from one frame to the next. For every logo it will calculate the usual visibility measures, such as exposure duration, on-screen coverage, size, and position. A saliency estimate taken at the location of the logo will then be combined with these measures to produce an attention-aware visibility score. Users can ask questions about the results in plain English and receive short summaries and reports.

2 Project Scope

The scope of this project includes the development of a sponsorship analytics application that can:

- Detect and track sponsor logos in sports broadcast footage.
- Compute visibility metrics such as exposure duration, coverage, size, and position.
- Compute an attention-aware visibility score using visual saliency.
- Answer questions and generate reports in plain English.
- Evaluate performance using metrics such as mean Average Precision (mAP), precision, and recall.

3 Overview of Developer's Responsibilities

The developers are responsible for:

- Designing and implementing the logo detection and tracking pipeline.
- Designing and implementing the attention-aware scoring method.
- Designing and implementing the natural language reporting layer.
- Creating a user-friendly interface for the application.
- Conducting thorough testing and documentation of the software.

4 Overall Description

4.1 Functionality of the Software

The software will allow users to:

- Load a sports broadcast video.
- Detect and track sponsor logos in the video.
- View visibility metrics and attention-aware scores for each brand.
- Ask questions and generate reports in plain English.
- Evaluate the performance of the detection and scoring process.

4.2 User Interaction

Users will interact with the application through a graphical user interface (GUI) that provides options for loading a video, running the analysis, viewing results, entering questions, and exporting reports.

4.3 Expected Outcomes and Performance Metrics

The expected outcomes include:

- Successful detection and tracking of sponsor logos.
- Accurate calculation of exposure metrics and attention-aware scores.
- Clear and accurate answers to questions about the results.
- Performance metrics such as mAP, precision, recall, and tracking accuracy will be used to evaluate the system.

5 Product Perspective

5.1 Standalone Application

The BEAMS application will be a standalone piece of software designed for personal computers. It will not require any external dependencies other than the specified libraries and models.

5.2 Utilisation of Libraries and Models

The application will use a YOLO object detector for finding logos, together with OpenCV and NumPy for image processing and calculation. A saliency model will estimate where viewers are likely to look, and LangGraph will handle the natural language reporting.

6 Product Functions

The primary functions of the product include:

- Logo detection in broadcast video frames.
- Logo tracking across frames.
- Calculation of exposure duration, coverage, size, and position.
- Calculation of the attention-aware visibility score.
- Question answering and report generation in plain English.

- Performance evaluation using metrics such as mAP, precision, and recall.

7 User Classes and Characteristics

The application will cater to the following user classes:

- Sponsors and Brand Managers: Users who wish to understand the value of their sponsorship exposure without needing technical expertise.
- Marketing Analysts: Users who compare brands and matches and prepare reports for clients.
- Event Organisers: Rights holders who need clear evidence of sponsor visibility when negotiating with sponsors.
- Researchers: Users who wish to evaluate the performance of the system.
- Developers: Individuals who may modify or extend the code for further development.

8 Operating Environment

8.1 Supported Operating Systems

The application will support the following operating systems:

- Windows
- macOS
- Linux

8.2 Hardware Requirements

The minimum hardware requirements for the application are:

- 8 GB RAM
- Multi-core processor
- GPU recommended

8.3 Software Requirements

The software requirements include:

- Python 3.x
- PyTorch
- OpenCV
- NumPy
- LangGraph

9 Design and Implementation Constraints

The design and implementation of the application will adhere to the following constraints:

- The scoring must combine visibility measures with a saliency-based attention estimate.
- The application must be developed using Python and compatible libraries.
- Answers given in plain English must be based on the computed results.

- The user interface must be intuitive and easy to navigate.

10 User Documentation

The user documentation will include:

- A user manual for installation and usage.
- Examples of video analysis and report generation.
- Troubleshooting and FAQs.

11 General Constraints

The application must adhere to the following general constraints:

- Detection quality must be maintained under motion blur, changing lighting, and partial occlusion.
- The application must support common video formats such as MP4 and AVI.

12 Assumptions and Dependencies

The following assumptions and dependencies apply:

- Users have basic knowledge of software operation.
- The required libraries, models, and a compatible operating system are available.
- Users have access to sports broadcast footage for analysis.
- Annotated logo datasets are available for training and evaluation.

13 Conclusion

13.1 Summary

This SRS document outlines the requirements for developing BEAMS, an application that detects and tracks sponsor logos in sports broadcasts, calculates an attention-aware visibility score, and reports the results in plain English.

13.2 Importance of the Proposed System

The proposed system is important for sponsorship measurement, especially in cases where screen time alone does not reflect the real value of an exposure.

13.3 Future Work and Potential Improvements

Future work may include:

- Supporting live broadcast analysis.
- Using video-based saliency models.
- Extending the system to more sports and logo types.
- Integrating the score into a sponsorship valuation model.